

Homework 4

Math 2031 — Professor Hamilton

Name: Imran Qasimi

Date: September 26, 2026

Section 5.3

Question 2

Statement: Prove that $n^2 - n$ is divisible by 2 for every integer n .

Proof:

Let $n \in \mathbb{Z}$. We want to show that $\forall n \in \mathbb{Z}, 2|n^2 - n$.

There are two possible cases.

A) n is odd, i.e, $n = 2k_1 + 1$

B) n is even, i.e, $n = 2k_2$

Case 1: Odd: $n = 2k_1 + 1$

Let n be an odd integer, thus $n = 2k_1 + 1$ for some integer k_1 .

We want to show that $2|n^2 - n$.

By assumption since n is odd we can rewrite our expression as:

$$(2k_1 + 1)^2 - (2k_1 + 1)$$

Which expands to:

$$4(k_1)^2 + 4k_1 + 1 - 2k_1 - 1$$

Simplifying:

$$4(k_1)^2 + 2k_1$$

Factoring out a 2:

$$2(2(k_1)^2 + k_1)$$

Thus if n is odd, $2|n^2 - n$.

Case 1: Even: $n = 2k_2$

Let n be an even integer, thus $n = 2k_2$ for some integer k_2 .

We want to show that $2|n^2 - n$.

By assumption $n = 2k_2$ thus we can rewrite our expression as:

$$(2k_2)^2 - (2k_2)$$

Which expands to:

$$4(k_2)^2 - 2k_2$$

Factoring out a 2:

$$2(2(k_2)^2 - k_2)$$

$$\therefore \forall n \in \mathbb{Z}, 2|n^2 - n.$$

■

Question 3

Statement: Prove that if $n \in \mathbb{Z}$, then $n^2 + 2$ is not divisible by 4 for every integer n .

Proof by Contradiction:

Suppose not. Suppose the negation: $n \in \mathbb{Z} \wedge 4|n^2 + 2$. However, $4 \nmid n^2 + 2$ for $n = 1$:

$$1^2 + 2 = 3$$

$$4 \nmid 3$$

Therefore since the negation is false, it must mean the original statement must be true.

■

Question 6c

Statement: If $a|b$ and $b|c$, then $a|c$.

Let $a, b, c \in \mathbb{N}$ such that $b = a(k_1)$ and $c = b(k_2)$.

We want to show that $a|c$, i.e, c is an integer multiple of a .

By assumption since $b = a(k_1)$ we can rewrite c as:

$$c = a(k_1)(k_2)$$

Thus we have shown $a|c$.

■